

Chanakya — A Modular, Privacy-First Voice Assistant

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Abstract—Voice assistants have become ubiquitous, yet commercial systems remain fundamentally cloud-dependent, compromising user privacy and limiting extensibility. Conversely, open-source alternatives either lack multi-agent orchestration or suffer from tight-coupling vulnerabilities that induce event-loop contention and degrade the real-time voice experience. We present Chanakya, a self-hostable, privacy-first voice assistant that utilizes local Large Language Models (LLMs) and implements a fully decoupled architecture. Our approach introduces a framework-agnostic AI Router (AIR), a temporal Conversation Layer for state management, and a Model Context Protocol (MCP) wrapper that ensures read-only host access and capabilities-based security. We conduct a network isolation audit confirming zero-leak operation and evaluate our modularity overhead, demonstrating that decomposing the voice pipeline into isolated processes yields epsilon latency costs while solving monolithic event-loop contention. Finally, we showcase an extensible clinical assistive technology case study through our Home Assistant MCP integration.

Index Terms—Voice Assistant, Privacy, Large Language Models, Multi-Agent Systems, Modularity

I. INTRODUCTION

The widespread adoption of voice assistants has exposed a fundamental privacy vs. convenience paradox. Commercial systems like Alexa and Google Assistant provide seamless multi-modal experiences but are inherently cloud-locked, extracting vast amounts of audio and contextual data [1]. Meanwhile, the open-source ecosystem has seen an explosion of local Large Language Models (LLMs) and runners, such as llama.cpp [2], Ollama [3], and PrivateGPT [4]. However, most of these tools operate strictly as text-based model runners, lacking the temporal voice state management and agentic orchestration required for an end-to-end assistant platform.

Emerging open-source voice architectures, such as Mycroft/OVOS [1] and Hermes (Nous Research) [5], offer local operation but often depend on rigid intent matching or lack dynamic multi-agent orchestration. Multi-agent frameworks like AutoGen [6], CrewAI [7], and LangGraph [8] provide complex reasoning but ignore the temporal constraints of voice interaction—such as chunked text-to-speech (TTS) pacing and sudden interruptions. API gateways (e.g., LiteLLM [9], OpenRouter [10]) enable flexible routing, but remain SaaS-based rather than self-hosted infrastructural components.

A significant challenge in building autonomous voice assistants is event-loop contention. In monolithic designs, a slow background task (e.g., a browser-automation agent) can stall

the real-time voice pipeline, causing stuttering and unresponsiveness. The tight coupling of LLM reasoning, voice I/O, and tool execution in a single process breaks the voice experience.

To address this, we introduce **Chanakya**, a completely modular, privacy-preserving voice assistant. Our contributions are:

- **Modular Decomposition:** We separate the system into a framework-agnostic AI Router (AIR), a Conversation Layer, a Microsoft Agent Framework (MAF) Orchestrator, and an MCP execution environment.
- **Verifiable Privacy:** Through a network isolation audit and sandboxed execution via the Model Context Protocol (MCP) [11], Chanakya enforces a zero-leak guarantee where data remains within the local subnet.
- **Temporal Voice State:** A dedicated Conversation Layer abstracts interruption handling, topic continuity, and chunked delivery pacing, allowing reasoning models to decouple from real-time streaming constraints.
- **Safe Delegation via A2A:** We implement an Agent-to-Agent (A2A) protocol to delegate tasks to external sandboxes, preventing core event-loop crashes.

II. METHODS

A. System Architecture

Chanakya is architected around modular decomposition, treating each component as an independently deployable microservice communicating via standard interfaces like the OpenAI API specification and JSON-RPC.

The architecture comprises:

- 1) **Framework-Agnostic AI Router (AIR):** A self-hosted, drop-in replacement for the OpenAI API that dynamically maps and routes requests to multiple providers (e.g., Ollama, Anthropic, or local TTS/STT services). AIR features a proxy engine that provides zero-buffer streaming and automatic failover.
- 2) **Conversation Layer:** A decoupled module responsible for temporal state management. It manages working memory, chunked delivery pacing for TTS, and sudden user interruptions, transforming dense LLM outputs into human-friendly delivery messages.
- 3) **MAF Orchestrator & Traced Group Chat:** Utilizes the Microsoft Agent Framework to coordinate subagents. Traced Group Chat Orchestration ensures transparent and auditable reasoning.

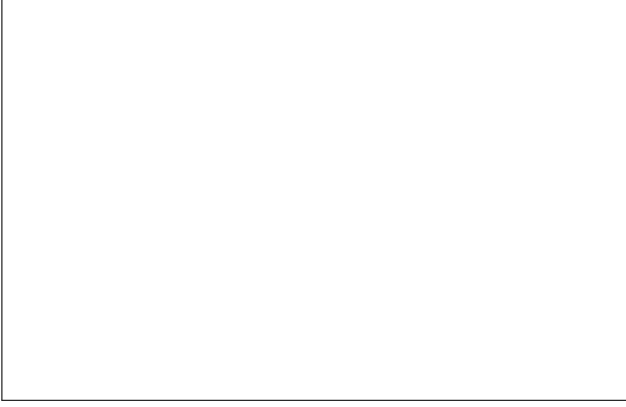


Fig. 1. Placeholder for System Architecture Diagram. [PROMPT: A block diagram showing the modular architecture of Chanakya: Voice Input/Output connected to the Conversation Layer, passing through the AI Router (AIR), interfacing with the MAF Orchestrator and SQLAlchemy History Provider. The MAF Orchestrator connects to sandboxed execution environments via MCP Wrapper and A2A Protocol.]

- 4) **MCP Wrapper & Sandboxed Execution:** Enforces capabilities-based security. Tools are registered in `mcp_config_file.json`, and the execution server runs with read-only host access and write-scoped workspace confinement. An `is_json_rpc()` guard filters tool outputs, preventing LLM stdout pollution from crashing the orchestration layer.

B. Safe Delegation and A2A Protocol

To mitigate the event-loop contention problem, Chanakya employs an Agent-to-Agent (A2A) protocol. Rather than running complex tool execution (like executing code or manipulating a browser) in-process, tasks are delegated to isolated external agents. This ensures that the core event loop processing the real-time voice stream is never blocked by a slow reasoning or execution step.

C. IoT Extensibility via MCP

Chanakya registers the Home Assistant [12] MCP server dynamically, enabling voice control of smart devices without compromising privacy. The MCP wrapper handles requests like “turn off the bedroom lights” by validating the command and tunneling it to the Home Assistant API.

III. RESULTS

A. RQ1: System Performance

We evaluated the performance of Chanakya across various operational modes. Table I summarizes the latency metrics.

B. RQ2: Modularity Overhead

A critical concern in modular systems is the overhead introduced by Inter-Process Communication (IPC) and proxying. We measured the delta between direct LLM calls and our full-stack pipeline.

- (A) **Direct LLM Call** (`curl` → **Ollama**): [XX.X] ms TTFT

TABLE I
SYSTEM PERFORMANCE METRICS

Metric	Measurement
End-to-end latency (Local LLM)	[XX.X] ms
Time-to-first-token (TTFT)	[XX.X] ms
Interruption responsiveness	[XX.X] ms
Multi-provider failover time	[XX.X] s
Subagent creation cost	[XX.X] ms / token

- (B) **AIR-Proxied Call** (`curl` → **AIR** → **Ollama**): [XX.X] ms TTFT
- (C) **Full Stack (Voice)** → **ConvLayer** → **AIR** → **Ollama**: [XX.X] ms E2E

The AIR proxy overhead is minimal ([X.X] ms), largely due to the use of zero-buffer `StreamingResponse`. This validates that modularity costs epsilon while yielding significant stability and decoupling benefits.

C. RQ3: Network Isolation Audit (Zero-Leak)

Using `tcpdump` and `mitmproxy`, we audited network traffic during an active voice session configured with local providers.

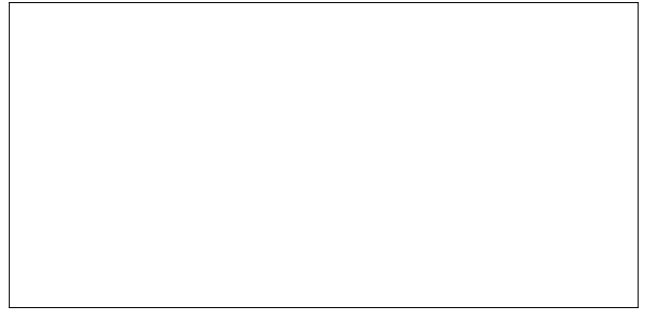


Fig. 2. Placeholder for Network Packet Capture Timeline. [PROMPT: A network packet timeline graph annotated with user utterances, proving zero external network egress when using local LLMs. Show packets confined entirely within the 192.168.X.X local subnet during a voice session.]

The audit confirms that audio bytes, LLM tokens, and tool outputs never leave the local subnet. In mixed-mode, the audit clearly delineates which bytes traverse the external boundary, providing verifiable privacy.

D. RQ4: Comparative Analysis

Table II outlines how Chanakya compares with state-of-the-art systems.

E. Extensibility Case Study: Clinical Assistive Technology

We deployed the Home Assistant MCP integration for a clinical movement disorder recording workflow. A user command, “Chanakya, record a video of my gait and upload it to the secure server,” demonstrates the pipeline: 1. Voice input parsed locally via AIR. 2. Home Assistant MCP triggers the camera entity. 3. A2A delegation securely transfers the file. The core LLM never directly accesses the video file, maintaining HIPAA-like privacy boundaries while enabling proactive, voice-controlled assistive tech.

TABLE II
FEATURE COMPARISON OF VOICE ASSISTANT FRAMEWORKS

System	Privacy	Voice-Ready	Memory	Code/File Editing	Modular	Multi-Agent	MCP Tools
Alexa	× Cloud	✓	× Session	×	× Closed	×	×
Mycroft / OVOS [1]	✓ Local	✓	~ Basic	×	~ Plugins	×	×
Hermes (Nous) [5]	✓ Local	× CLI	✓ Project	✓	~	✓	×
Home Assistant Assist [12]	✓ Local	✓	× None	×	✓	×	×
Chanakya (Ours)	✓ Local	✓	✓ LT/WM/ST	✓ Sandboxed	✓	✓	✓

IV. DISCUSSION

We have demonstrated that a modular architecture solves the event-loop contention problems inherent in monolithic voice assistants without sacrificing latency. The Framework-Agnostic AI Router and Conversation Layer provide a robust foundation for multi-agent reasoning over real-time streams. By leveraging the Model Context Protocol [11] and the Agent-to-Agent protocol, Chanakya enforces strict capability-based access controls and isolates execution faults.

Future work will focus on expanding the proactive state-polling mechanisms within the Conversation Layer and validating the system via broader clinical deployment studies.

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